NCERT Solutions for Class 12

Biology

Chapter 12 -Ecosystem

1. Fill in the blanks.

(a) Plants are called as_____ because they fix carbon dioxide.

Ans: Plants are called autotrophs because they fix carbon dioxide.

(b) In an ecosystem dominated by trees, the pyramid (of numbers) is _____ type.

Ans: In an ecosystem dominated by trees, the pyramid (of numbers) is of inverted type.

(c) In aquatic ecosystems, the limiting factor for productivity is _____.

Ans: In aquatic ecosystems, the limiting factor for productivity is light.

(d) Common detritivores in our ecosystem are_____.

Ans: Common detritivores in our ecosystem are earthworms.

(e) The major reservoir of carbon on earth is_____.
Ans: A major reservoir of carbon on Earth is oceans.

2. Which one of the following has the largest population in a food chain?

(a) Producers

(b) Primary consumers

(c) Secondary consumers

(d) Decomposers

Ans: (d) Decomposers

Decomposers include microorganisms like bacteria and fungi. A largest population is formed by them in a food chain and they obtain nutrients by breaking down the remains of dead plants and animals.

3. The second trophic level in a lake is

(a) Phytoplankton

(b) Zooplankton

(c) Benthos

(d) Fishes

Ans: (b) Zooplankton

Primary consumers are at the level of Second trophic. These are the organisms that eat the primary producers. Zooplankton is also one of them.

They are the primary consumers in aquatic food chains that feed upon phytoplankton and efficiently digest plant matter for energy.

4. Secondary producers are

(a) Herbivores

(b) Producers

(c) Carnivores

(d) None of the above

Ans: (d) None of the above.

Plants are the only producers and they lie at the lowest trophic level. Thus, they are called primary producers. There are no other producers in the food chain.

5. What is the percentage of photosynthetically active radiation (PAR), in the incident solar radiation.

(a) 100%

(b) 50 %

(c) 1-5%

(d) 2-10%

Ans: (b) 50%

About fifty percent of total incident solar radiation forms photosynthetically active radiation of PAR.

6. Distinguish between

(a) Grazing food chain and detritus food chain

Ans:

Grazing food chain	Detritus food chain
Energy is derived from the Sun in the grazing food chain.	Energy comes from organic matter which is generated in trophic levels of the grazing foodchain.
It starts with the producers, available at the first trophic level.	It starts with detritus like dead bodies of animals or fallen leaves,

The plant biomass is then eaten by herbivores, which successively are consumed by a spread of carnivores.	which are then consumed by decomposers or detritivores. These detritivores are in turn eaten by their predators.
This food chain is a large one.	Detritus food chain is smaller as compared to the grazing food chain.

(b) Production and decomposition

Ans:

Production	Decomposition
It is the rate of organic matter produced by producers.	Breaking down of complex organic matter with the help of decomposers from the body of dead plants and animals and converting into organic raw materials like CO_2, H_2O, and other nutrients.
It is dependent on the photosynthetic capacity of the producers.	It occurs with the help of decomposers.
The requirement of sunlight is necessary by plants for primary production.	No sunlight requirement is there in the process of decomposition.

(c) The upright and inverted pyramid

Ans:

Upright pyramid	Inverted pyramid
The pyramid of energy is always upright.	The pyramid of biomass and the a pyramid of numbers can be inverted.
The number and biomass of organisms in the producer level of an ecosystem are the highest in an upright pyramid, which further keeps on decreasing at each trophic level in a food chain.	The number and biomass of organisms in the producer level of an ecosystem are the lowest in an inverted pyramid, which further keeps on increasing at each trophic level.

(d) Food chain and Food web

Ans:

Food chain	Food chain
It is a single linear sequence of organisms.	It is a network formed by interconnecting food chains.
In this chain, members are present at higher trophic levels which sustain single types of organisms.	Alternate food sources are present in one organism.

(e) Litter and detritus

Ans:

Litter	Detritus
Litter is all the wastes that are produced above the ground.	Detritus is the remains of dead plants and animals in an

	ecosystem.
Litter consists of both biodegradable and non-biodegradable matter.	Detritus consists of only biodegradable matter.

(f) Primary and secondary productivity

Ans:

Primary productivity	Secondary productivity
It is defined as the rate at which organic energy is stored as organic matter produced by producers per unit an area over a while.	It is defined as the rate of production of organic matter by consumers over a while.

7. Describe the components of an ecosystem.

Ans: An ecosystem creates a link that has both the biological community as well as the non-living components of an area. The ecosystem was defined as the living world and its habitat by A.G. Tansley. It meant that any habitat on the earth which includes plants, animals, and non-living substances interact to produce materials that can be exchanged between living and nonliving components of the habitat in an ecosystem. Thus, they perform as a unit, which gets conspicuous during the processes of nutrient cycling, energy flow, decomposition, and productivity. Many ecosystems like ponds, forests, grasslands, etc are there.

The two components of an ecosystem are:

- Biotic component: The living component of an ecosystem is the biotic component. Which includes biotic factors such as producers, consumers,

decomposers, etc. It is divided into major groups:

(a) Producers are green plants that include plants and algae. The presence of chlorophyll pigment, helps them carry out the process of photosynthesis in the presence of light. Thus, they are also referred to as converters or transducers.

(b) Consumers or heterotrophs are organisms that cannot manufacture food and are, therefore, directly (primary consumers) or indirectly (secondary and tertiary consumers) dependent on producers for their food.

(c) Decomposers are the microorganisms like bacteria and fungi. The largest population is contributed by them in a food chain by breaking down the remains of dead plants and animals, and nutrients are obtained.

- Abiotic component: These are the nonliving components within an ecosystem such as light, temperature, water, soil, air, inorganic nutrients, etc.

8. Define ecological pyramids and describe with examples, pyramids of number and biomass.

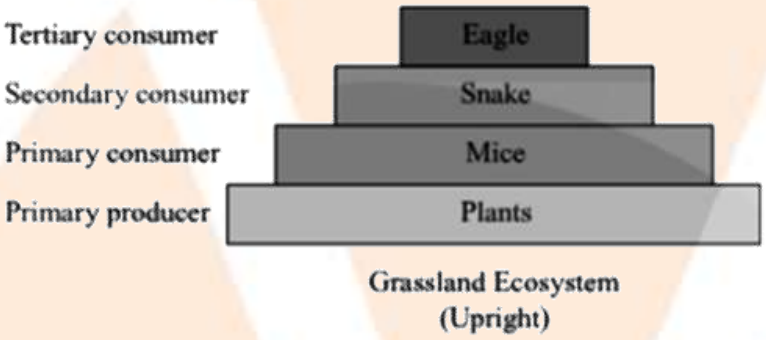
Ans: An ecological pyramid is a graphical representation of the number, biomass, and energy of the successive trophic levels of an ecosystem. In an ecological pyramid, the lowermost trophic level is formed by the producers and the topmost level is that of carnivores.

There are three types of pyramids:

- Pyramid of numbers
- Pyramid of energy
- Pyramid of biomass

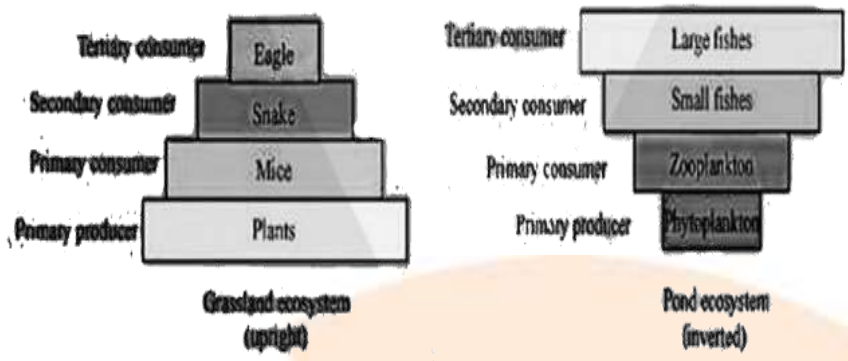
Pyramid of numbers: It shows the association between the number of producers, herbivores, and carnivores. Depending on the number of

producers, it can be upright or inverted. In a grassland ecosystem, grasses are the producers. The pyramid of numbers is often upright. In this type of food chain, the number of producers (plants) is followed by the number of herbivores (mice), which in turn is followed by the number of secondary consumers (snakes) and tertiary carnivores (eagles). Hence, At the producer level the number of individuals will be greatest, while at the top carnivores, the number of individuals present will be least.



Whereas on the other hand, in a parasitic food chain, the pyramid of numbers is inverted. In this type of food chain, several fruit-eating birds are provided with food by a single tree (producer), which in turn support several insect species.

Pyramid of biomass: A pyramid of biomass may be a graphical representation of the whole amount of living matter present at each trophic level of an ecosystem. It can be upright or inverted. The pyramid of biomass in grasslands and forest ecosystems is upright as the amount of biomass present at the producer level is higher than at the top carnivore level. In a pond ecosystem, the pyramid of biomass is inverted as the biomass of fishes far exceeds the biomass of zooplankton (upon which they feed).



9. What is primary productivity? Give a brief description of factors that affect primary productivity.

Ans: It is defined as the amount of organic matter or biomass produced by producers per unit area over a while. The primary productivity of an ecosystem depends on a variety of environmental factors such as light, temperature, water, precipitation, etc. It also depends on the availability of nutrients and the availability of plants to carry out photosynthesis.

10. Define decomposition and describe the processes and products of decomposition.

Ans: Decomposition is the process that involves the breakdown of complex organic matter or biomass with the help of decomposers from the body of dead plants and animals into inorganic raw materials like carbon dioxide, water, and other nutrients. The processes involved in decomposition are as follows:

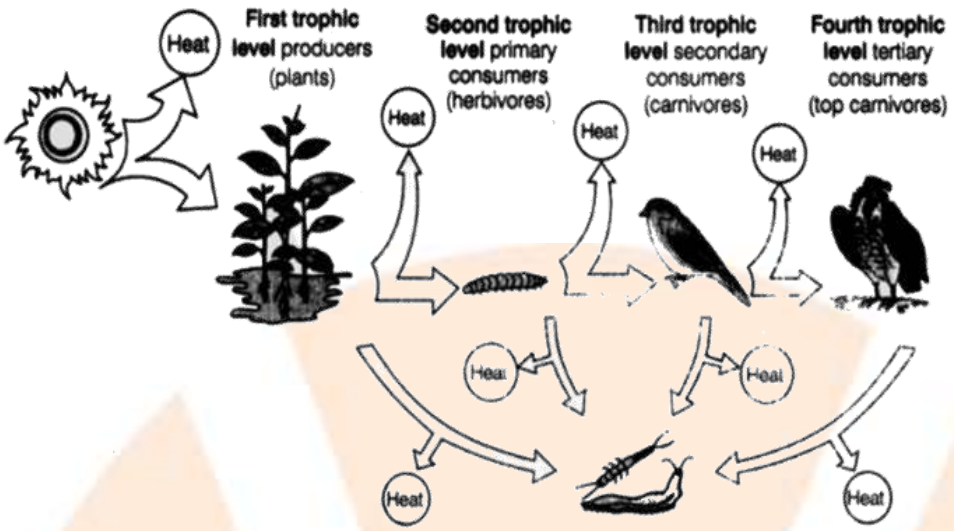
- **Fragmentation:** The first step in the process of decomposition is fragmentation. By the action of detritivores, there is the breakdown of detritus into smaller pieces like in earthworms is called fragmentation.
- **Leaching:** Several water-soluble inorganic substances percolate into the deeper layers of soil along with rain or irrigation water, it is known as leaching.

- **Catabolism:** Bacteria and fungi degrade detritus through various enzymes into smaller pieces is called catabolism.
- **Humification:** The formation of a finely divided, amorphous dark-colored colloidal substance called humus, which acts as a reservoir of nutrients for plants is called humification.
- **Mineralization:** By the action of microbes, humus is further degraded which finally results in the release of inorganic nutrients into the soil.

  A dark-colored, nutrient-rich substance called humus is produced by the process of decomposition. Humus thus degrades and releases inorganic raw materials such as CO_2, water, and other nutrients in the soil.

11. Give an account of energy flow in an ecosystem.

Ans: In ecosystems, useful energy flows in a one-way path. It enters the living things from the physical world, passes from one organism to a different and eventually escapes back to the physical environment in a less useful form. Energy enters an ecosystem from the Sun. Solar radiations are absorbed by the Earth's surface through the atmosphere. The process of photosynthesis is carried out by these radiations. Also, they help to maintain the Earth's temperature for the survival of living organisms. Few of the solar radiations are reflected by the Earth's surface. However, a very little 2-10 percent of solar energy is captured by green plants (producers) during photosynthesis to be converted into food. The total rate of photosynthesis, including the organic matter that is almost immediately used up in plant respiration, is termed as 'gross primary productivity. When these green plants are consumed by herbivores, only 10% of the stored energy from producers is transferred to herbivores. For various processes such as respiration, growth, and reproduction, the remaining 90% of the energy is used. Similarly, only 10% of the energy of herbivores is transferred to carnivores. Therefore, this is called the ten percent law of energy flow.



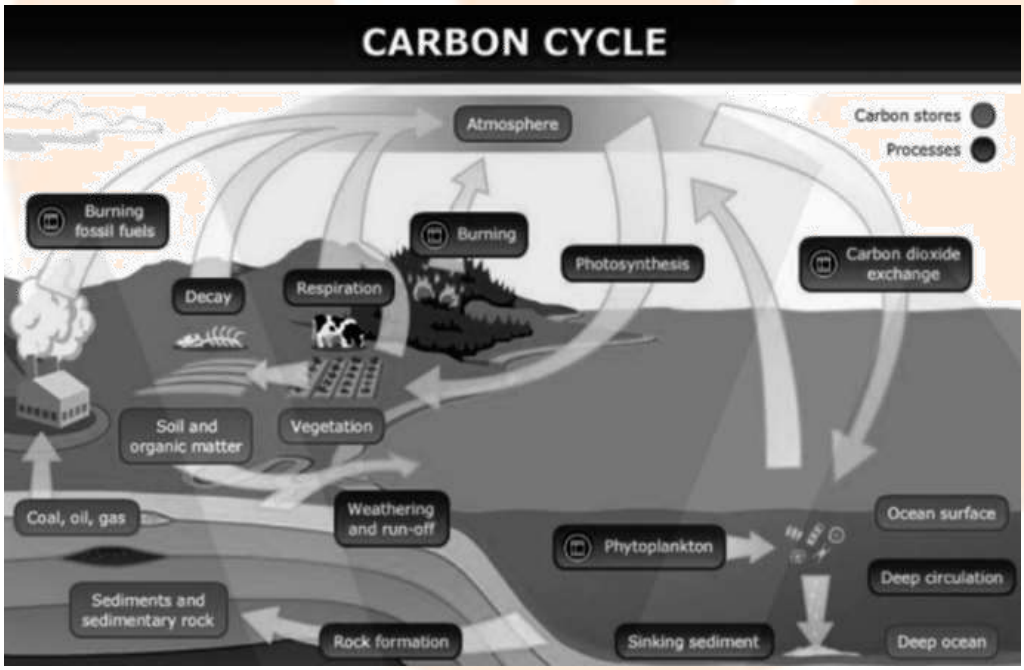
Diagrammatic representation of flow of energy through different trophic levels

12. Write important features of a sedimentary cycle in an ecosystem?

Ans: Sedimentary cycles have their reservoir pool in deep sediments or the earth's crust or rocks. Nutrient elements that are found in the sediments of the Earth are Sulphur, phosphorus, potassium, and calcium. There is an absence of a gaseous phase in this cycle.

This cycle is very slow. A longer time is taken to complete their circulation and is considered as less perfect cycles. Since, In the process of recycling, nutrient elements get locked in the reservoir pool, which thereby takes a long time to come out and continue circulation. Thus, it always goes out of circulation for an extended time.

13. Outline salient features of carbon cycling in an ecosystem.

Ans: An important gaseous cycle is the carbon cycle which has its reservoir pool in the atmosphere. Organic components of all organisms always have a carbon major body constituent. It is considered to be a fundamental element found in all living forms. All molecules like carbohydrates, lipids, and proteins required for life processes are made of carbon. Carbon is present in

combined form in all living forms by a fundamental process called ‘photosynthesis. Sunlight and atmospheric carbon dioxide are used to produce a carbon compound called ‘glucose’ in the process of photosynthesis. cycleliving organisms also utilize this glucose molecule. Thus, atmospheric carbon gets inside in living forms. Now, it becomes important to recycle this absorbed carbon dioxide back into the atmosphere to complete the cycle. There are several processes by which carbon is recycled back into the atmosphere in the form of carbon dioxide gas. First is the process of respiration in which glucose molecules are broken to produce carbon dioxide gas. Second is the process of decomposition which also releases carbon dioxide from dead bodies of plants and animals into the atmosphere. Industrialization, Combustion of fuels deforestation, volcanic eruptions, and forest fires act as other paramount sources of carbon dioxide.